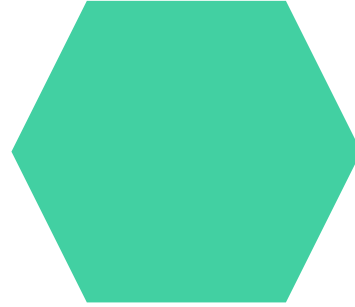




Seasonal Agriculture Performance Analysis

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Major Data Analytics Project | VOIS

Internship Program



PROBLEM STATEMENT

- **Agricultural Context:** Agricultural productivity in India is governed by distinct seasons (Kharif, Rabi, Zaid), resulting in dramatic variations in rainfall, temperature, humidity, and soil moisture.
- **Core Industry Challenge:** Raw agricultural data does not clearly explain how farm management practices, resource usage (water, fertilizers, pesticides), and economic outcomes interact across seasonal cycles.
- **Primary Objective:** Analyze 4,000 farm records across 8 states to investigate seasonal performance disparities, identify key yield drivers, assess water efficiency, and map disease/pest vulnerabilities.
- **Key Questions Addressed:** Which season offers the highest return on investment? How do specific crops perform across seasons? Are observed differences across seasons statistically significant?



Project Description

Dataset Granularity:

Evaluates 4,000 farm records across 8 Indian States (Maharashtra, Karnataka, Andhra Pradesh, Gujarat, Punjab, Tamil Nadu, Telangana, Madhya Pradesh) spanning 28 physical, operational, and financial features.

Three-Season Analytical Framework:

- Kharif (Monsoon): High precipitation (avg 849 mm), peak moisture (31.1%), dominant output (5.63 t/ha), elevated pest risk (54.5%).
- Rabi (Winter): Favorable cool climate (avg 23.5°C), stable productivity (5.04 t/ha), dependable cash margins.
- Zaid (Summer): Extreme heat (avg 31.0°C), low rainfall (305 mm), peak water usage (6,420 m³), net deficits.

End-to-End Analytics Workflow:

Data Quality Auditing -> Median Imputation -> 1.5xIQR Outlier Assessment -> Univariate/Bivariate/Multivariate EDA -> Non-Parametric Hypothesis Testing (Kruskal-Wallis, Chi-Square, Spearman) -> Decision Recommendations.

WHO ARE THE END USERS?

- **1. Farmers & Farmer Producer Organizations (FPOs):** Enables data-driven seasonal crop selection to maximize profit margins (e.g. Chilli/Sugarcane) and avoids water-intensive cereal crops during water-deficit Zaid months.
- **2. Agricultural Extension Officers & Agronomists:** Provides empirical evidence to advocate for precision micro-irrigation (Drip/Sprinkler) and proactive pest control before monsoon humidity surges.
- **3. Policy Makers & Ministry of Agriculture:** Directs targeted irrigation and power subsidies to water-stressed regions, and informs MSP procurement planning based on seasonal supply forecasts.
- **4. Agri-Fintech, Banks & Crop Insurance Providers:** Improves agricultural credit scoring, loan risk calibration, and premium pricing based on seasonal farm profitability dynamics.

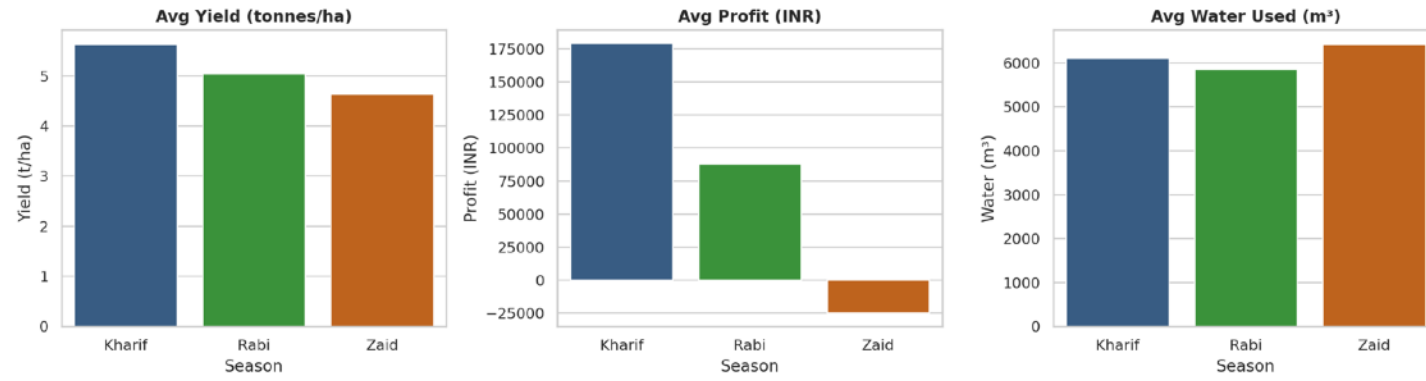
Technology Used

- **Programming Language:** Python 3.10 (High performance numerical execution & data science ecosystem)
- **Data Manipulation & Profiling:** Pandas & NumPy (DataFrames, GroupBy aggregations, missing value median imputation, IQR boundary modeling)
- **Exploratory Data Visualization:** Matplotlib & Seaborn (Statistical heatmaps, grouped bar charts, KDE histograms, boxplots, pairplots)
- **Inferential & Hypothesis Testing:** SciPy.stats (Kruskal-Wallis non-parametric H-tests, Chi-Square contingency analysis, Spearman rank correlation)
- **Environment & Version Control:** Google Colab, Jupyter Notebooks, Git & GitHub for reproducible open-source research



RESULTS: Executive Summary & Seasonal Dynamics

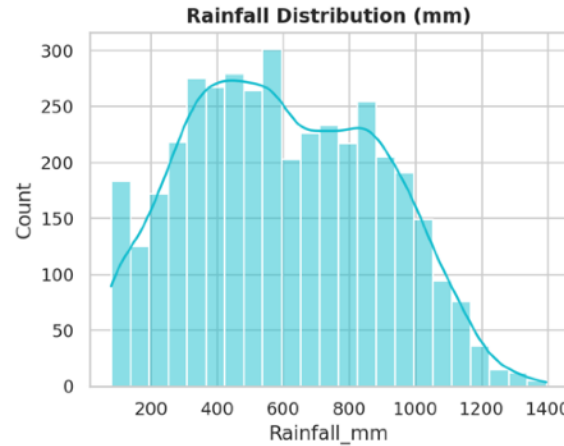
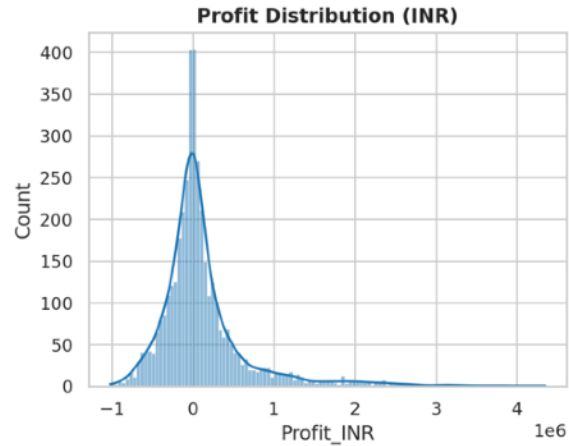
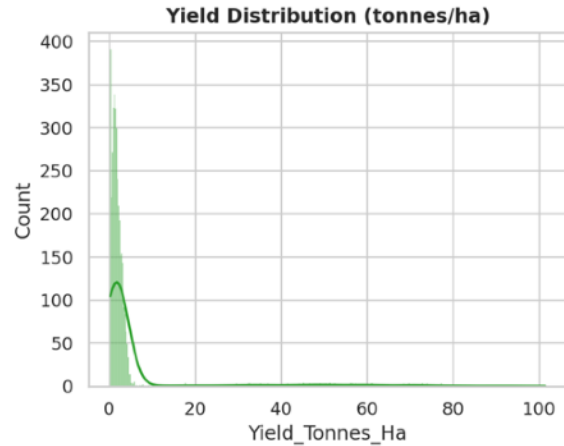
Seasonal Agricultural Performance Overview (4,000 Farms)



- **Kharif (Peak Output):** Generates highest average yield (5.63 t/ha) and net profits (INR 178,914/farm; 51.5% profitable farms) supported by 849.2 mm monsoon rainfall.
- **Rabi (Seasonal Stability):** Steady productivity (5.04 t/ha) and positive margins (INR 87,689/farm) driven by controlled 23.5°C winter climate and reliable soil moisture.
- **Zaid (Deficit Period):** Financial stress phase (-INR 24,804/farm) with highest water consumption (6,419.9 m³) due to severe heat (31.0°C) and low rainfall (304.7 mm).
- **Water Efficiency:** Natural precipitation boosts Kharif water efficiency (5.89 t/1,000 m³), whereas Zaid yields lower output per unit volume of water.

RESULTS: Exploratory Data Analysis & Distributions

Distributions of Key Agricultural & Economic Variables

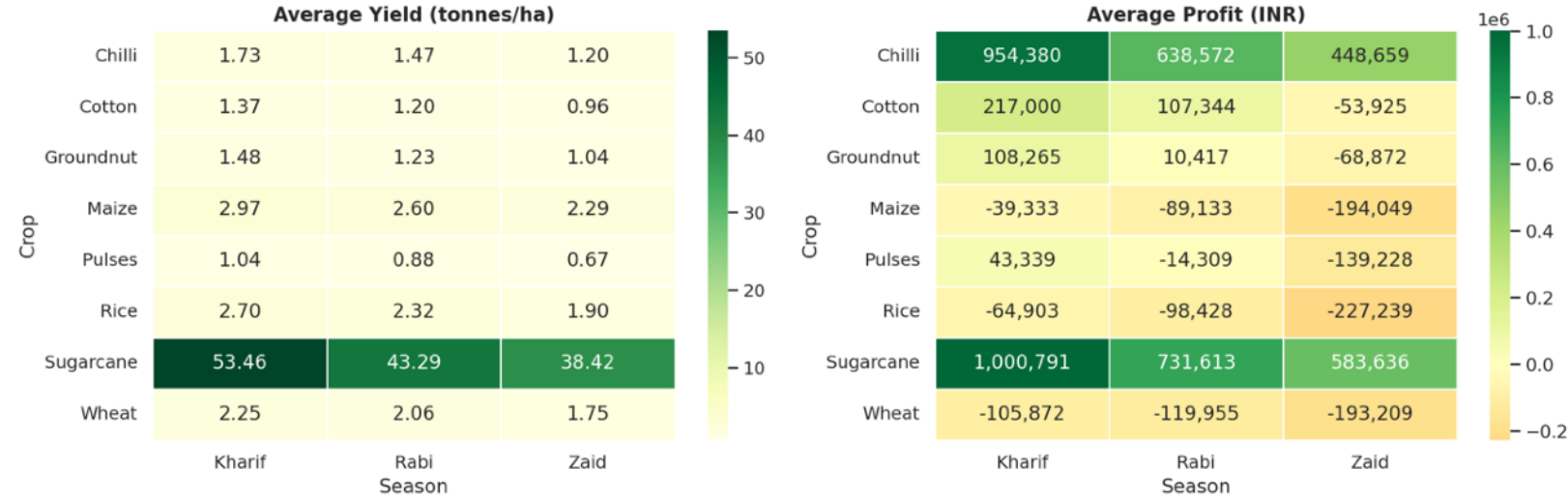


- **Data Quality & Cleaning:** Missingness in Rainfall (1.2%), Soil Moisture (1.0%), and Yield (0.8%) successfully resolved via median imputation. Zero duplicates detected across 4,000 rows.
- **Skewed Yield Profile:** Agricultural yield is right-skewed (median: 1.74 t/ha vs mean: 5.24 t/ha) due to massive biomass production in Sugarcane (up to 101.4 t/ha).

- **Profit Distribution:** Farm profit ranges from -INR 1.02M to +INR 4.35M (median: INR 3,673), demonstrating wide economic variance across crop types and seasonal windows.
- **Environmental Consistency:** Rainfall shows multi-modal distribution aligned with monsoon patterns, while Temperature and Humidity are normally distributed.

RESULTS: Crop-Season Interactions & Heatmaps

Crop Performance Dimensions Across Seasons



- **Sugarcane Dominance:** Achieves highest physical tonnage across all seasons (Kharif: 53.46 t/ha, Rabi: 43.29 t/ha, Zaid: 38.42 t/ha) and net profits exceeding INR 1.0M.
- **High-Margin Chilli:** Commands superior farm profit (Kharif: INR 954k, Rabi: INR 638k, Zaid: INR 448k) despite modest yield (~1.5 t/ha), benefiting from premium market price (>INR 90,000/tonne).

- **Cereal Deficits (Rice & Wheat):** Both staple crops suffer negative net profits across multiple seasons (-INR 64k to -INR 227k) due to high input costs relative to market realizations.
- **Water Efficiency Matrix:** Sugarcane achieves peak physical efficiency (36.0 t/1,000 m³ in Kharif), whereas Rice requires substantial water volumes for modest physical output.

RESULTS: Statistical Hypothesis Testing

Rigorous non-parametric statistical tests were executed to validate that seasonal differences are genuine and statistically significant:

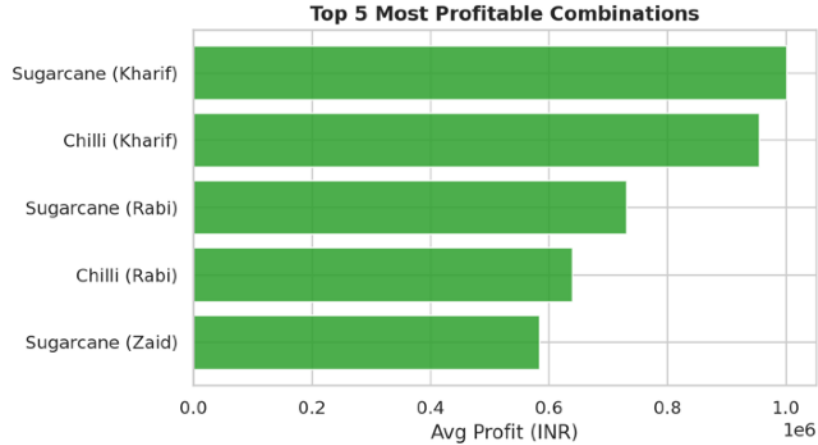
- **Yield across Seasons (Kruskal-Wallis):** $H = 29.41$, $p = 4.11e-07$ -> Statistically Significant. Yield distributions differ significantly across Kharif, Rabi, and Zaid.
- **Profit across Seasons (Kruskal-Wallis):** $H = 58.83$, $p = 1.68e-13$ -> Highly Significant. Farm profit is strongly season-dependent.
- **Irrigation Method Independence (Chi-Square):** $\chi^2 = 2.45$, $p = 0.8738$ -> Fail to Reject. Current irrigation method choices do not vary by season, indicating clear modernization opportunities.

Formal Statistical Hypothesis Testing Results Summary

Hypothesis Test	Variables Tested	Test Statistic	p-value	Decision (alpha=0.05)
Kruskal-Wallis H-Test	Yield across Seasons	$H = 29.41$	$p = 4.11e-07$	Statistically Significant
Kruskal-Wallis H-Test	Profit across Seasons	$H = 58.83$	$p = 1.68e-13$	Statistically Significant
Kruskal-Wallis H-Test	Water Usage across Seasons	$H = 14.22$	$p = 8.16e-04$	Statistically Significant
Chi-Square Test	Season vs. Irrigation Method	$\chi^2 = 2.45$ (df=6)	$p = 0.8738$	Fail to Reject (Independent)
Spearman Correlation	Rainfall vs. Yield	$\rho = 0.082$	$p = 1.90e-07$	Statistically Significant

RESULTS: Profitability Extremes & Risk Assessment

Crop-Season Profitability Extremes



- **Top 5 Profitable Combinations:** Sugarcane (Kharif: INR 1.00M), Chilli (Kharif: INR 954k), Sugarcane (Rabi: INR 732k), Chilli (Rabi: INR 639k), Sugarcane (Zaid: INR 584k).

- **Bottom 5 Deficit Combinations:** Rice (Zaid: -INR 227k), Maize (Zaid: -INR 194k), Wheat (Zaid: -INR 193k), Pulses (Zaid: -INR 139k), Wheat (Rabi: -INR 120k).



- **Disease & Pest Vulnerability:** Pest risk is highest during Kharif (54.5%) due to monsoon humidity (71.8%) and elevated soil moisture (31.1%), dropping to 38.2% in Zaid.

- **Cost vs Revenue Trade-Off:** Average costs remain high (~INR 513k to 544k) across all seasons, but Kharif revenues (INR 711k) vastly surpass Zaid revenues (INR 519k).

Future Scope

1. Predictive Machine Learning Modeling:

Develop supervised machine learning pipelines (Random Forest, XGBoost, CatBoost) to forecast crop yields and compute profit probabilities before the sowing window.

2. IoT Sensor Integration & Precision Agriculture:

Deploy field IoT sensor networks to continuously monitor soil moisture, micro-climates, and N-P-K nutrient depletion for automated, precision drip irrigation scheduling.

3. Satellite Remote Sensing & NDVI Tracking:

Integrate multi-spectral satellite imagery (Sentinel-2 / Landsat) to track vegetation indices (NDVI/EVI), detecting crop stress, drought severity, and pest hotspots in real-time.

4. Dynamic Digital Advisory & Market Platform (DSS):

Build a bilingual mobile-first decision support application offering localized sowing recommendations, weather risk alerts, and real-time mandi market price trends to eliminate intermediary exploitation.

GitHub Repository

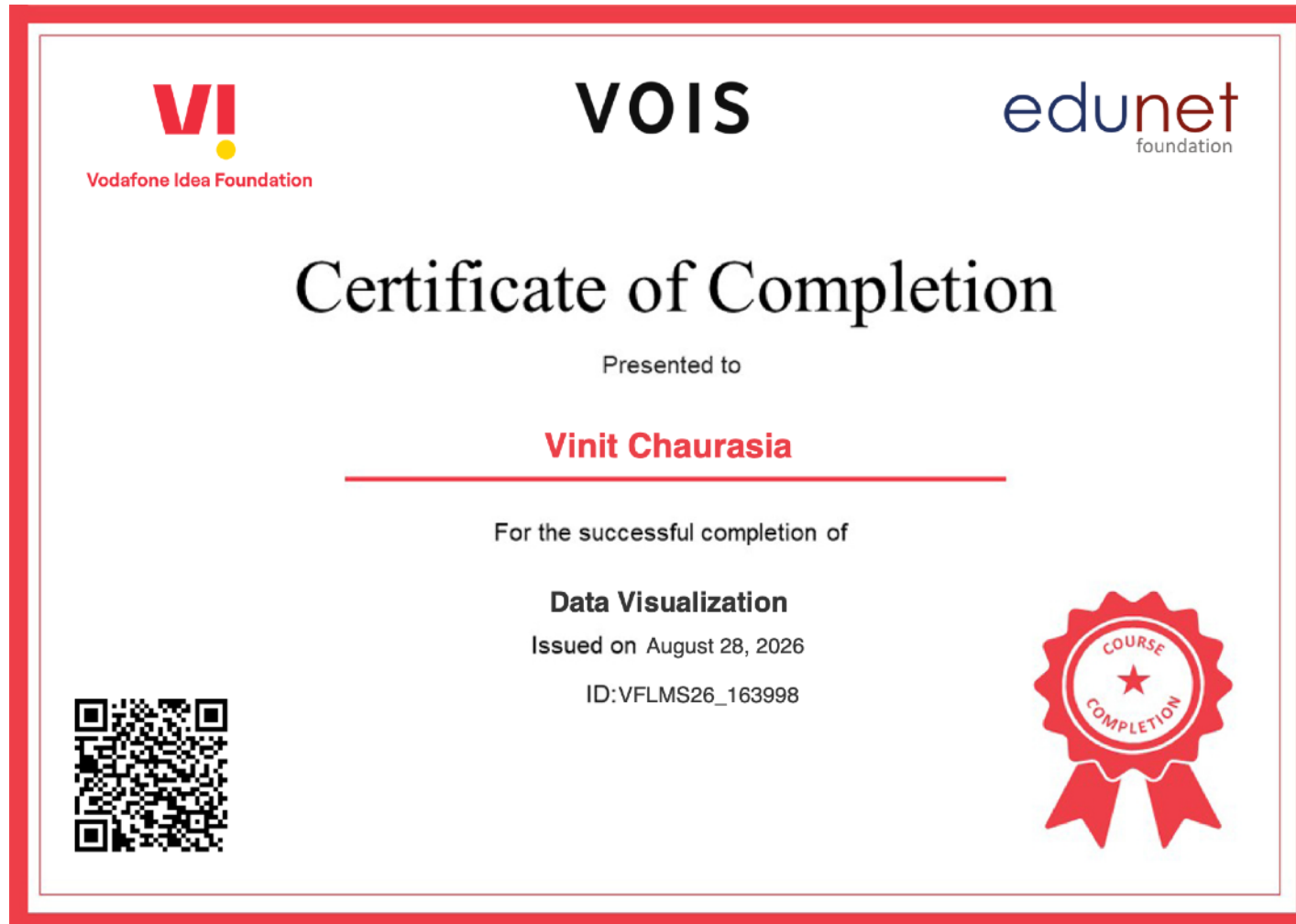
Repository Link:

https://github.com/chourasiavinit9-dev/Seasonal_Major_project

Repository Contents:

- Seasonal_Agriculture_Performance_Data_Analytics.ipynb (Complete, clean Colab/Jupyter Notebook)
- seasonal_agriculture_performance_dataset.csv (Cleaned & imputed 4,000-record dataset)
- High-resolution visualization exports, correlation matrices & statistical testing scripts
- Comprehensive project documentation, methodology reports, and rubric compliance verification

VOIS Course Completion Certificate



Course Name: Data Visualization

Thank You!

Seasonal Agriculture Performance Analysis

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Open for Questions & Discussion